Task 3: SQL Data Exploration and Analysis Using Dataset 2

Step 1: Database and Table Setup

Objective: To create a separate database and verify the table structures for the spacecraft communication dataset.

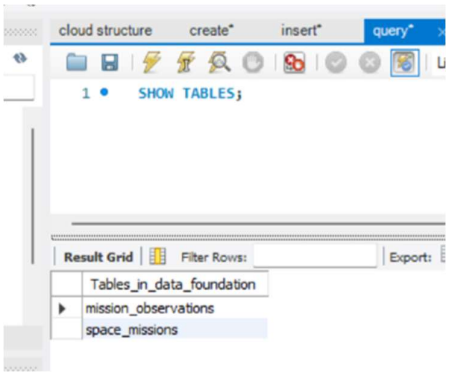
Database Name: space_communications

Tables Used:

1. spacecraft — stores spacecraft names, targets, agencies, launch dates, and statuses.
2. communication_logs — stores ground-station details, communication dates, signal strengths, and transmission durations.

SQL Concepts Used: CREATE DATABASE, USE, CREATE TABLE, SHOW TABLES, DESCRIBE, primary keys, and foreign keys.

Observation: The database and table structures were created and checked in MySQL Workbench. The spacecraft_id foreign key connects communication records to their corresponding spacecraft.



Step 2: Data Loading and Verification

Objective: To populate the spacecraft and communication-log tables and verify that the records were inserted correctly.

SQL Concepts Used: INSERT INTO, VALUES, SELECT, and COUNT().

Data Loading Performed:

- Inserted 8 spacecraft records into the spacecraft table.
- Inserted 10 communication records into the communication_logs table.
- Retrieved the inserted records and checked the total row counts.

Result: The database contains 8 spacecraft records and 10 communication-log records.

```
USE space_communications;

SELECT * FROM spacecraft;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap C

spacecraft_id	spacecraft_name	target	agency	launch_date	status
S001	Mars Observer	Mars	NASA	2023-01-15	Active
S002	Lunar Explorer	Moon	ISRO	2022-08-10	Active
S003	Voyager Probe	Interstellar Space	NASA	1977-09-05	Active
S004	Europa Clipper	Europa	NASA	2024-10-14	In Transit
S005	Chandrayaan Mission	Moon	ISRO	2023-07-14	Completed
S006	Saturn Surveyor	Saturn	ESA	2021-06-22	Active
S007	Solar Research Probe	Sun	NASA	2020-02-09	Active
S008	Neptune Pathfinder	Neptune	ESA	2024-03-18	Planned

```
SELECT * FROM communication_logs;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conter

log_id	spacecraft_id	ground_station	communication_date	signal_strength	transmission_minutes
201	S001	Madrid Ground Station	2024-01-12	82.456	14.50
202	S001	Canberra Ground Station	2024-02-18	79.234	12.80
203	S002	Bengaluru Ground Station	2024-03-05	91.678	18.20
204	S003	Goldstone Ground Station	2024-03-22	45.123	6.40
205	S004	Madrid Ground Station	2024-11-01	76.891	22.70
206	S004	Canberra Ground Station	2024-11-15	74.567	20.30
207	S005	Bengaluru Ground Station	2024-04-09	94.321	16.90
208	S006	Madrid Ground Station	2024-05-21	68.456	11.60
209	S007	Goldstone Ground Station	2024-06-30	88.765	19.40
210	S002	Bengaluru Ground Station	2024-07-17	90.234	17.10

Step 3 :

Query 1: Pattern-Based Filtering and Result Limiting

Objective: To identify spacecraft whose names contain the word "Probe" and display the matching records in descending launch-date order.

SQL Concepts Used: LIKE, WHERE, ORDER BY DESC, and LIMIT.

Purpose: The query searches for spacecraft names containing a specified text pattern, sorts the matching records by launch date, and limits the output to five records.

```
SELECT
    spacecraft_id,
    spacecraft_name,
    target,
    agency,
    launch_date
FROM spacecraft
WHERE spacecraft_name LIKE '%Probe%'
ORDER BY launch_date DESC
LIMIT 5;
```

esult Grid | Filter Rows: | Edit: | Export/Import

spacecraft_id	spacecraft_name	target	agency	launch_date
S007	Solar Research Probe	Sun	NASA	2020-02-09
S003	Voyager Probe	Interstellar Space	NASA	1977-09-05

Query 2: Range-Based Filtering

Objective: To retrieve communication records whose signal strength falls between 75 and 90.

SQL Concepts Used: WHERE, BETWEEN, and ORDER BY DESC.

Purpose: The query filters communication logs using a specified signal-strength range and sorts the matching records from highest to lowest signal strength.

```
2   SELECT
3       log_id,
4       spacecraft_id,
5       ground_station,
6       communication_date,
7       signal_strength,
8       transmission_minutes
9   FROM communication_logs
10  WHERE signal_strength BETWEEN 75 AND 90
11  ORDER BY signal_strength DESC;
12
```

esult Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conter

log_id	spacecraft_id	ground_station	communication_date	signal_strength	transmission_minutes
209	S007	Goldstone Ground Station	2024-06-30	88.765	19.40
201	S001	Madrid Ground Station	2024-01-12	82.456	14.50
202	S001	Canberra Ground Station	2024-02-18	79.234	12.80
205	S004	Madrid Ground Station	2024-11-01	76.891	22.70
NULL	NULL	NULL	NULL	NULL	NULL

Query 3: Combining Related Tables Using LEFT JOIN

Objective: To display all spacecraft along with their communication records, including spacecraft without any communication logs.

SQL Concepts Used: LEFT JOIN, ON, table aliases, and ORDER BY ASC.

Purpose: The query combines the spacecraft and communication_logs tables using the common spacecraft_id column. Unlike an INNER JOIN, a LEFT JOIN retains every spacecraft from the left table, even when no matching communication record exists.

```
2   SELECT
3       s.spacecraft_id,
4       s.spacecraft_name,
5       s.agency,
6       c.log_id,
7       c.ground_station,
8       c.signal_strength,
9       c.transmission_minutes
10  FROM spacecraft AS s
11  LEFT JOIN communication_logs AS c
12      ON s.spacecraft_id = c.spacecraft_id
13  ORDER BY s.spacecraft_id ASC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

log_id	spacecraft_id	ground_station	communication_date	signal_strength	transmission_minutes
209	S007	Goldstone Ground Station	2024-06-30	88.765	19.40
201	S001	Madrid Ground Station	2024-01-12	82.456	14.50
202	S001	Canberra Ground Station	2024-02-18	79.234	12.80
205	S004	Madrid Ground Station	2024-11-01	76.891	22.70
NULL	NULL	NULL	NULL	NULL	NULL

Query 4: Rounded Aggregate Analysis

Objective: To compare average communication signal strength across space agencies.

SQL Concepts Used: LEFT JOIN, COUNT(), AVG(), ROUND(), GROUP BY, and ORDER BY DESC.

Purpose: The query joins spacecraft details with communication records, groups the results by agency, counts the communication records, calculates the average signal strength, rounds the average to two decimal places, and sorts the agencies by average signal strength.

```
2   SELECT
3       s.agency,
4       COUNT(c.log_id) AS total_communications,
5       ROUND(AVG(c.signal_strength), 2) AS avg_signal_strength
6   FROM spacecraft AS s
7   LEFT JOIN communication_logs AS c
8       ON s.spacecraft_id = c.spacecraft_id
9   GROUP BY s.agency
10  ORDER BY avg_signal_strength DESC;
11
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content

log_id	spacecraft_id	ground_station	communication_date	signal_strength	transmission_minutes
209	S007	Goldstone Ground Station	2024-06-30	88.765	19.40
201	S001	Madrid Ground Station	2024-01-12	82.456	14.50
202	S001	Canberra Ground Station	2024-02-18	79.234	12.80
205	S004	Madrid Ground Station	2024-11-01	76.891	22.70
NULL	NULL	NULL	NULL	NULL	NULL

Conclusion :

The spacecraft communication dataset was stored in a separate MySQL database using two related tables: spacecraft and communication_logs. Four SQL queries were implemented to explore and analyse the data.

The queries demonstrated pattern-based filtering using LIKE, sorting and limiting results using ORDER BY and LIMIT, range-based filtering using BETWEEN, combining related tables using LEFT JOIN, and calculating rounded averages using AVG(), ROUND(), and GROUP BY.

These operations helped identify spacecraft matching a text pattern, filter communication records by signal strength, display spacecraft without communication logs, and compare average signal strength across space agencies.

Overall, the task provided practical experience in SQL filtering, joins, aggregation, and interpreting structured communication data.